

SET 9 – Online Learning Activity Analysis

AIM

To analyze online learning activity using R programming by creating a histogram, boxplot, scatter plot, and monthly trend line chart with moving average smoothing.

ALGORITHM

1. Import the Online Learning Activity dataset.
2. Create a histogram of Quiz Score.
3. Create a boxplot of Quiz Score by Course.
4. Create a scatter plot between Study Time and Quiz Score.
5. Represent Videos Watched using bubble size and apply transparency.
6. Convert Login Date into Date format.
7. Calculate average Quiz Score for each month.
8. Plot the monthly average using a line chart.
9. Apply moving average smoothing and identify the trend.

R PROGRAM

```
library(ggplot2)
```

```
data <- data.frame(  
  Student_ID=c("L01","L02","L03","L04","L05","L06"),  
  Gender=c("Male","Female","Male","Female","Male","Female"),  
  Age=c(20,22,19,21,23,20),  
  Course=c("R","R","SQL","R","R","SQL"),  
  Study_Time=c(3.5,4.2,2.0,5.0,2.5,4.0),  
  Videos_Watched=c(12,15,8,18,9,14),  
  Quiz_Score=c(78,85,65,92,70,88),  
  Login_Date=as.Date(c("2025-01-05","2025-01-05","2025-02-08",  
    "2025-02-08","2025-03-12","2025-03-12"))  
)  
  
ggplot(data,aes(Quiz_Score))+  
  geom_histogram(binwidth=5,fill="skyblue",color="black")+  
  labs(title="Distribution of Quiz Scores",x="Quiz Score",y="Frequency")
```

```
ggplot(data,aes(Course,Quiz_Score,fill=Course))+  
geom_boxplot()+  
labs(title="Quiz Score by Course",x="Course",y="Quiz Score")
```

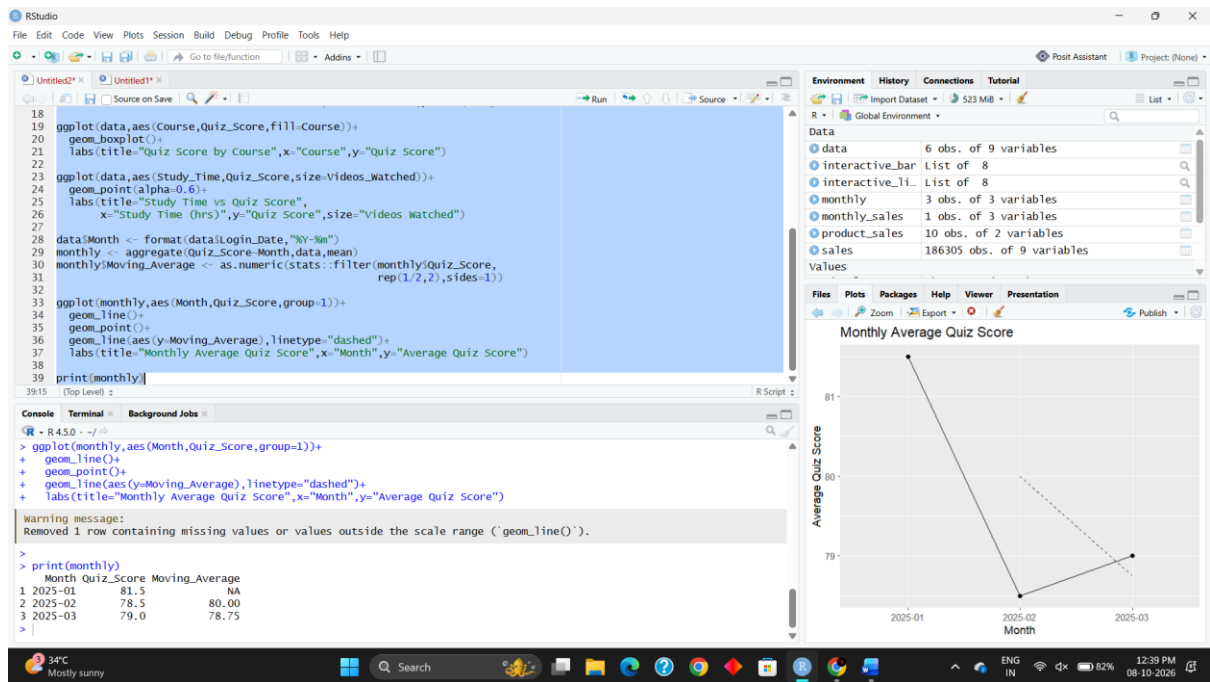
```
ggplot(data,aes(Study_Time,Quiz_Score,size=Videos_Watched))+  
geom_point(alpha=0.6)+  
labs(title="Study Time vs Quiz Score",  
x="Study Time (hrs)",y="Quiz Score",size="Videos Watched")
```

```
data$Month <- format(data$Login_Date,"%Y-%m")  
monthly <- aggregate(Quiz_Score~Month,data,mean)  
monthly$Moving_Average <- as.numeric(stats::filter(monthly$Quiz_Score,  
rep(1/2,2),sides=1))
```

```
ggplot(monthly,aes(Month,Quiz_Score,group=1))+  
geom_line()+  
geom_point()+  
geom_line(aes(y=Moving_Average),linetype="dashed")+  
labs(title="Monthly Average Quiz Score",x="Month",y="Average Quiz Score")
```

```
print(monthly)
```

OUTPUT:



RESULT

The histogram shows the distribution of quiz scores. The boxplot compares performance between R and SQL courses. The scatter plot indicates that students with higher study time generally achieve higher quiz scores. Monthly average scores were approximately **81.5, 78.5 and 79.0**, showing a slight decrease followed by a stable trend.

SET 10 – Survey Responses Analysis

AIM

To analyze survey responses using R programming by creating grouped bar charts, stacked bar charts, and a table of survey responses.

ALGORITHM

1. Create the survey response dataset.
2. Count the responses for Question 1.
3. Create a grouped bar chart for Question 1.
4. Convert the data into long format.
5. Count responses for all three questions.
6. Create a stacked bar chart.
7. Display the complete survey response table.
8. Analyze the response distribution.

R PROGRAM

```
library(ggplot2)
```

```

survey <- data.frame(
  Survey_ID=1:3,
  Question_1=c("A","B","C"),
  Question_2=c("B","A","A"),
  Question_3=c("C","D","B")
)

```

```

q1 <- as.data.frame(table(survey$Question_1))
names(q1) <- c("Answer","Count")

```

```

ggplot(q1,aes(Answer,Count,fill=Answer))+
  geom_bar(stat="identity")+
  labs(title="Grouped Bar Chart - Question 1",
  x="Answer",y="Count")

```

```

long <- reshape(
  survey,
  varying=c("Question_1","Question_2","Question_3"),
  v.names="Answer",
  timevar="Question",
  times=c("Question 1","Question 2","Question 3"),
  direction="long"
)

```

```

counts <- as.data.frame(table(long$Question,long$Answer))
names(counts) <- c("Question","Answer","Count")

```

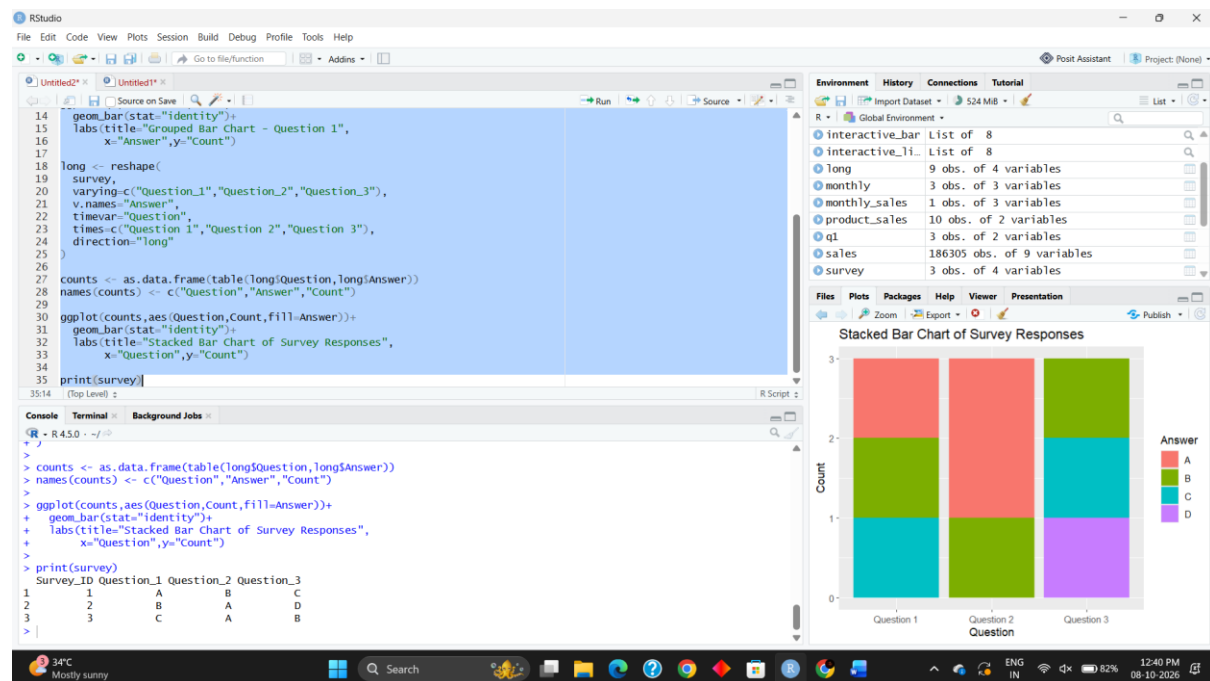
```

ggplot(counts,aes(Question,Count,fill=Answer))+
  geom_bar(stat="identity")+
  labs(title="Stacked Bar Chart of Survey Responses",
  x="Question",y="Count")

```

```
print(survey)
```

OUTPUT:



RESULT

The grouped bar chart displays the distribution of answers for Question 1. The stacked bar chart shows the overall distribution of responses for all three questions. The survey table successfully displays the responses for each survey participant.

SET 11 – Product Category Analysis

AIM

To analyze product category sales using R programming by creating a pie chart, funnel-style chart, and sales table.

ALGORITHM

1. Create the product category sales dataset.
2. Calculate the total sales.
3. Create a pie chart to show sales distribution.
4. Create a funnel-style chart using sales values.
5. Display the category-wise sales table.
6. Compare sales among the product categories.

R PROGRAM

```
library(ggplot2)
```

```
sales <- data.frame(
  Category=c("Electronics","Clothing","Appliances"),
  Sales=c(50000,35000,40000)
)
```

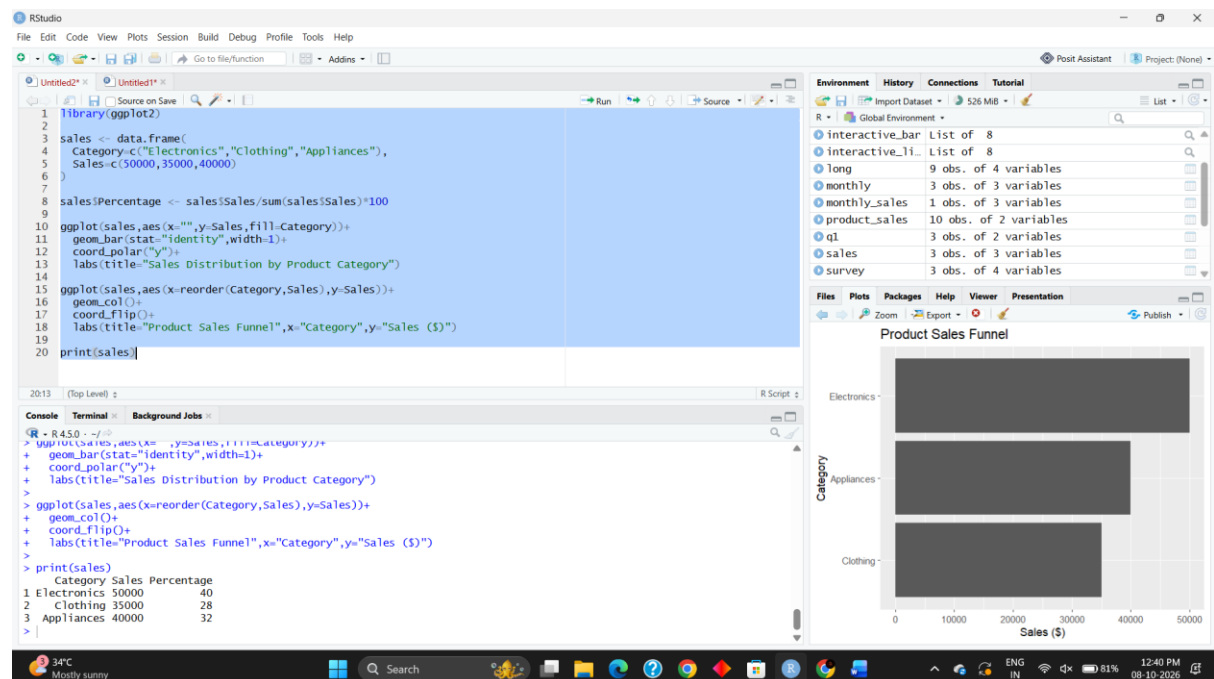
```
sales$Percentage <- sales$Sales/sum(sales$Sales)*100
```

```
ggplot(sales,aes(x="",y=Sales,fill=Category))+
  geom_bar(stat="identity",width=1)+
  coord_polar("y")+
  labs(title="Sales Distribution by Product Category")
```

```
ggplot(sales,aes(x=reorder(Category,Sales),y=Sales))+
  geom_col()+
  coord_flip()+
  labs(title="Product Sales Funnel",x="Category",y="Sales ($)")
```

```
print(sales)
```

OUTPUT:



RESULT

The pie chart shows that **Electronics has the highest sales (\$50,000)**, followed by **Appliances (\$40,000)** and **Clothing (\$35,000)**. The total sales are **\$125,000**. The charts and table successfully represent the product category sales.

SET 12 – Website Traffic Analysis

AIM

To analyze website traffic using R programming by creating a line chart for page views, a bar chart for top click-through rates, and a table of daily traffic data.

ALGORITHM

1. Create the website traffic dataset.
2. Convert Date into Date format.
3. Create a line chart for daily page views.
4. Sort the data according to click-through rate.
5. Select the top N days.
6. Create a bar chart for the selected days.
7. Display the website traffic table.
8. Analyze the website traffic trend.

R PROGRAM

```
library(ggplot2)
```

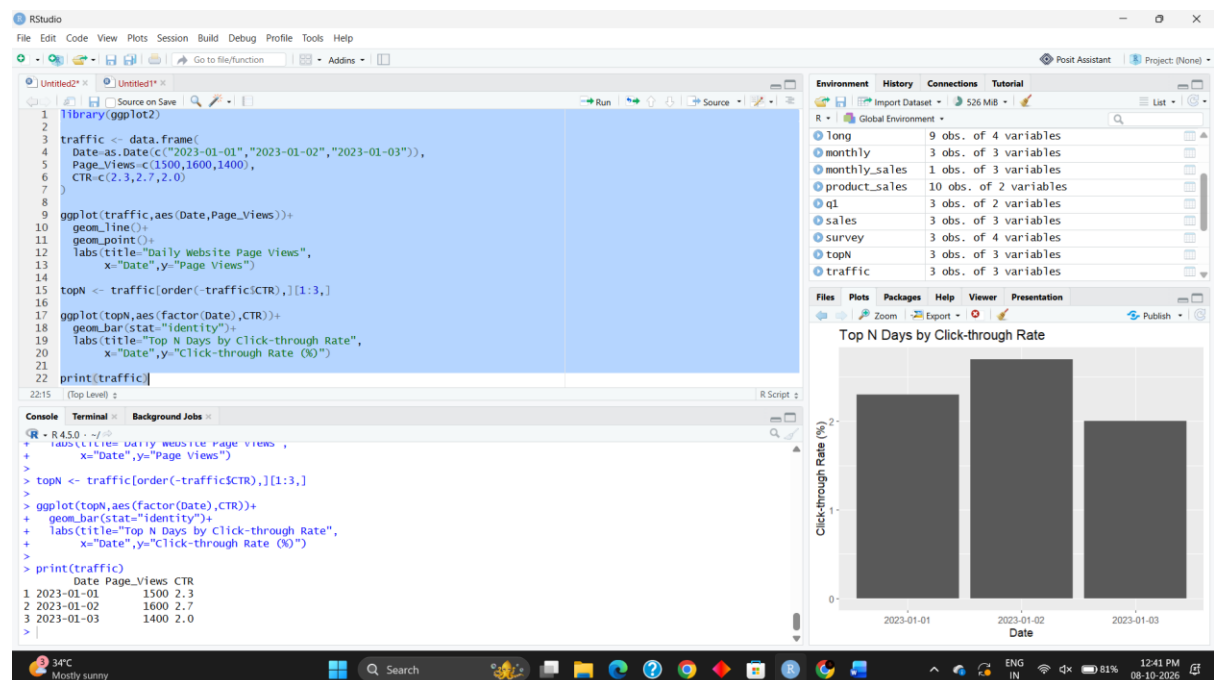
```
traffic <- data.frame(  
  Date=as.Date(c("2023-01-01","2023-01-02","2023-01-03")),  
  Page_Views=c(1500,1600,1400),  
  CTR=c(2.3,2.7,2.0)  
)
```

```
ggplot(traffic,aes(Date,Page_Views))+  
  geom_line()+  
  geom_point()+  
  labs(title="Daily Website Page Views",  
        x="Date",y="Page Views")
```

```
topN <- traffic[order(-traffic$CTR),][1:3,]
```

```
ggplot(topN,aes(factor(Date),CTR))+  
geom_bar(stat="identity")+  
labs(title="Top N Days by Click-through Rate",  
x="Date",y="Click-through Rate (%)")  
  
print(traffic)
```

OUTPUT:



RESULT

The line chart shows that page views increased from **1500 to 1600** and then decreased to **1400**. The highest click-through rate is **2.7% on 2023-01-02**. The website traffic data is successfully displayed using charts and a table.